Final Exam

● Graded

Student

Ruoying Wang

Total Points

44 / 45 pts

Question 1

Overall Grades **44** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 1 pt** Point adjustment

MATH 202-001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Ruoying Wang	rw355	Ruoying Wang

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: Ruoying Wang

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	4	5	5	5	5	5	5	5	44

Problem 1. [5] Find the general solution of the linear system:

$$\begin{aligned} 2x - 3y &= 1 \\ -2y - 3z &= 1 \\ x + 2z &= 1 \end{aligned}$$

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The equations can be written into the form of augment matrix $[A|b]$.

$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right]$ The question requires the general solution of $Ax=b$.

Therefore. we need to calculate both particular solution and homogeneous solution.

For $Ax=0$ $\left[\begin{array}{ccc|c} 2 & -3 & 0 & 0 \\ 0 & -2 & -3 & 0 \\ 1 & 0 & 2 & 0 \end{array}\right]$ $R_3 \leftarrow R_1 - 2R_3$ $\left[\begin{array}{ccc|c} 0 & -3 & -4 & 0 \\ 0 & -2 & -3 & 0 \\ 1 & 0 & 2 & 0 \end{array}\right]$ $R_1 \leftarrow -2R_1$ $\left[\begin{array}{ccc|c} 0 & 6 & 8 & 0 \\ 0 & -2 & -3 & 0 \\ 1 & 0 & 2 & 0 \end{array}\right]$

$R_2 \leftarrow \frac{1}{2}R_2$ $\left[\begin{array}{ccc|c} 0 & 6 & 8 & 0 \\ 0 & -6 & -9 & 0 \\ 1 & 0 & 2 & 0 \end{array}\right]$ $R_2 \leftarrow R_2 + R_1$ $\left[\begin{array}{ccc|c} 0 & 6 & 8 & 0 \\ 0 & 0 & -1 & 0 \\ 1 & 0 & 2 & 0 \end{array}\right]$ $\cdots$ $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$.

$\det(A) = 2\times(-4) + (-3)\times(-1)$
$\times(0+3)$
$= 1 \neq 0.$

Since A is full rank. there is no free variable in the solution.

$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -5 \end{array}\right]$ $\begin{array}{l} x = 11 \\ y = 7 \\ z = -5. \end{array}$

General solution: $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 11 \\ 7 \\ -5 \end{pmatrix}$

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Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$ Find $[121] = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix}$

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

(1) By casting-out algorithm. we can eliminate A into REF form. and select pivot columns from original column space since these columns are linearly independent.

$\begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \xrightarrow{R_2 \leftarrow -\frac{1}{5}R_2} \begin{bmatrix} 2 & 4 & 2 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 + R_3} \begin{bmatrix} 2 & 4 & 2 \\ 1 & 3 & 2 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \xrightarrow{R_1 \leftarrow R_4 - R_1} \begin{bmatrix} 1 & 4 & 2 \\ 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftarrow \frac{1}{2}R_1} \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$\xrightarrow{R_1 \leftarrow R_1 + R_3} \begin{bmatrix} 1 & 3 & 2 \\ 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1 - 3R_2} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\dim(\text{Col}(A)) = 2$

basis of $\text{Col}(A)$: $\left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$

(2) We can eliminate A into REF form and select non-zero rows in REF form. since they are linearly independent. $\dim(\text{Row}(A)) = 2$

Therefore. basis of $\text{Row}(A)$: $\left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}^T, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}^T \right\}$ −0.5

(3). To find basis of null space of A is to find basis vector for $Ax = 0$.

$\left[\begin{array}{ccc|c} 2 & 4 & 2 & 0 \\ -5 & 0 & 5 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 3 & 2 & 0 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ $V = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

$\dim(\text{Null}(A)) = 1$.

what is the dimension of V ?

basis of $\text{Null}(A)$: $\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$

−0.5

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad T\left(\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \qquad T\left(\begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = A \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

Since T is linear transformation $\mathbb{R}^3 \to \mathbb{R}^2$. $A \in \mathbb{R}^{2\times 3}$.
Let e_1, e_2, e_3 be the standard basis vectors respectively $e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ $e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ $e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$
Define A as $\begin{bmatrix} | & | & | \\ T(e_1) & T(e_2) & T(e_3) \\ | & | & | \end{bmatrix}$.

$$\begin{cases} T(e_2) - T(e_3) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ T(e_1) - T(e_3) = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \\ -T(e_3) = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \end{cases} \qquad \begin{cases} T(e_1) = \begin{bmatrix} -2 \\ 0 \end{bmatrix} \\ T(e_2) = \begin{bmatrix} -1 \\ -1 \end{bmatrix} \\ T(e_3) = \begin{bmatrix} -2 \\ -2 \end{bmatrix} \end{cases}$$

$$A = \begin{bmatrix} -2 & -1 & -2 \\ 0 & -1 & -2 \end{bmatrix}.$$

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Problem 4. [5] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$

To find Jordan canonical form of A, we need to find eigenvalues.

$\det(A-\lambda I) = 0$

$A-\lambda I = \begin{bmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{bmatrix}$ $\quad \det(A-\lambda I) = (1-\lambda)^2(2-\lambda)$.

$\lambda = 1 \;/\; 2$.

$\lambda = 1$ $(A-I)V = 0$ $\left[\begin{array}{ccc|c} 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & -2 & 0 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ $V_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. $\quad (A-I)V_2 = V_1$

$V_2 = \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 0 \end{bmatrix}$

GM = 1

AM = 2.

$\lambda = 2$ $(A-2I)V = 0$ $\left[\begin{array}{ccc|c} -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -2 & -2 & -1 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 0 & 6 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{6} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$

$V_3 = \begin{bmatrix} -2 \\ -1 \\ 6 \end{bmatrix}$

GM = 1

AM = 1.

$J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ $\quad P = \begin{bmatrix} 0 & -\frac{1}{2} & -2 \\ 0 & 0 & -1 \\ 1 & 0 & 6 \end{bmatrix}$ $\quad P^{-1} = \begin{bmatrix} 0 & 6 & 1 \\ -2 & +4 & 0 \\ 0 & -1 & 0 \end{bmatrix}$

$A = PJP^{-1}$

$= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & -\frac{1}{2} & -2 \\ 0 & 0 & -1 \\ 1 & 0 & 6 \end{bmatrix} \begin{bmatrix} 0 & 6 & 1 \\ -2 & 4 & 0 \\ 0 & -1 & 0 \end{bmatrix}$

$= \begin{bmatrix} 0 & -\frac{1}{2} & -\frac{4}{2} \\ 0 & 0 & -1 \\ 1 & 1 & 12 \end{bmatrix} \begin{bmatrix} 0 & 6 & 1 \\ -2 & 4 & 0 \\ 0 & -1 & 0 \end{bmatrix}$

$= \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$

Problem 5. [5] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$f(x,y,z) = \vec{x} A \vec{x}^T$ let $\vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}^T$

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$$

$\det(A-\lambda I) = 0$.

$$A - \lambda I = \begin{bmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{bmatrix}$$

$$\begin{aligned} \det(A-\lambda I) &= (3-\lambda)((\lambda-3)(\lambda-4)-1) + \\ &\quad -(3-\lambda) \\ &= (3-\lambda)(\lambda-3)(\lambda-4) - 2) \\ &= -(\lambda-3)(\lambda-5)(\lambda-2) \end{aligned}$$

For symmetric matrix A. there exist basis which is orthonormal.
For each eigenvalue. the corresponding eigenvectors are linearly independent.
Therefore. the three eigen vectors can be a basis.

$\lambda = 3$ $(A-3I)X = 0$.

$$\left[\begin{array}{ccc|c} 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ 1 & -1 & 1 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad V_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad u_1 = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \\ 0 \end{bmatrix}$$

$\lambda = 2$. $(A-2I)X = 0$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & -1 & 2 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad V_2 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \quad u_2 = \begin{bmatrix} -\frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{bmatrix}$$

$\lambda = 5$ $(A-5I)X = 0$

$$\left[\begin{array}{ccc|c} -2 & 0 & 1 & 0 \\ 0 & -2 & -1 & 0 \\ 1 & -1 & -1 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & -\frac{1}{2} & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad V_3 = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix} \quad u_3 = \begin{bmatrix} \frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{3} \end{bmatrix}$$

$$Q = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{6} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} & -\frac{\sqrt{6}}{6} \\ 0 & -\frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{3} \end{bmatrix}$$

$u_1 \cdot u_2 = 0$
$u_2 \cdot u_3 = 0$
$u_1 \cdot u_3 = 0$.

Therefore. Q is an orthonormal basis matrix

$$Q^T = Q^{-1} = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ -\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} \\ \frac{\sqrt{6}}{6} & -\frac{\sqrt{6}}{6} & \frac{\sqrt{6}}{3} \end{bmatrix}$$

$$D = Q^T A Q = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$A = QDQ^T = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{6} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} & -\frac{\sqrt{6}}{6} \\ 0 & \frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{3} \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ -\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} \\ \frac{\sqrt{6}}{6} & -\frac{\sqrt{6}}{6} & \frac{\sqrt{6}}{3} \end{bmatrix}$$

Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

To find a, b, c. define $A = [v_1, v_2, v_3]$ $x = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$.

$Ax = w$.

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 + R_3} \left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 0 & 1 & 1 & 0 \\ -2 & 2 & 0 & -8 \end{array}\right] \xrightarrow{R_3 \leftarrow R_1 + R_2} \left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 0 & 3 & 0 & -3 \\ -2 & 2 & 0 & -8 \end{array}\right]$$

$$\xrightarrow{R_2 \leftarrow \frac{1}{3}R_2} \left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 0 & 1 & 0 & -1 \\ -2 & 2 & 0 & -8 \end{array}\right] \xrightarrow{R_3 \leftarrow -\frac{1}{2}R_3} \left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 0 & 1 & 0 & -1 \\ 1 & -1 & 0 & 4 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 + R_2} \left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 3 \end{array}\right]$$

$$\xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left[\begin{array}{ccc|c} 0 & 0 & -1 & -1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 3 \end{array}\right] \xrightarrow{R_1 \leftarrow -R_1} \left[\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 3 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$$

Problem 7. [5'] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V. orthogonal + normalize.

$\begin{bmatrix}0&3\\1&0\\1&2\\0&-1\end{bmatrix} \sim \begin{bmatrix}1&0\\0&1\\0&0\\0&0\end{bmatrix}$ $\dim(V) = 2$. The basis of V needs 2 vectors.

$v_1 = u_1 \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$

By gram-schmidt algorithm

$v_2 = u_2 - \frac{u_2 \cdot v_1}{\|v_1\|^2} v_1$

$= \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{2}{2}\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$

$= \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$

$v_1 \cdot v_2 = 0 - 1 + 1 + 0 = 0$, $v_1 \perp v_2$.

$v_1' = \begin{bmatrix}0\\\frac{\sqrt{2}}{2}\\\frac{\sqrt{2}}{2}\\0\end{bmatrix}$ $v_2' = \begin{bmatrix}\frac{\sqrt{3}}{3}\\-\frac{\sqrt{3}}{6}\\\frac{\sqrt{3}}{6}\\-\frac{\sqrt{3}}{6}\end{bmatrix}$

A orthonormal basis for V:

$$\left\{\begin{bmatrix}0\\\frac{\sqrt{2}}{2}\\\frac{\sqrt{2}}{2}\\0\end{bmatrix}, \begin{bmatrix}\frac{\sqrt{3}}{3}\\-\frac{\sqrt{3}}{6}\\\frac{\sqrt{3}}{6}\\-\frac{\sqrt{3}}{6}\end{bmatrix}\right\}$$

Problem 8. [5'] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

~~To prove that $\det A = 0$, we need to find $Ax = 0$ doesn't only have trivial solution. Assume $Ax = 0$.~~

~~$x^T A x = (A^T x)^T x = (-Ax)^T x$~~

Since $A^T = -A$.

$\det(A^T) = \det(-A)$

$\det(A^T) = \det(A)$.

$\det(A) = \det(-A)$. $\det(-A) = (-1)^n \det(A)$.

~~If n is odd~~ If n is odd, $\det(-A) = -\det(A)$.

$\det(A) = -\det(A)$

Therefore $\det(A) = 0$. The proof holds.

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Problem 9. [5] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other. $\vec{u}\cdot\vec{v}=0$. $\vec{u}\perp\vec{v}$.

Given $\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|$

$(\|\vec{u}+\vec{v}\|)^2 = (\|\vec{u}-\vec{v}\|)^2$

$\|\vec{u}\|^2 + 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2 = -2\vec{u}\cdot\vec{v} + \|\vec{u}\|^2 + \|\vec{v}\|^2$

$\vec{u}\cdot\vec{v} = 0.$

Therefore $\vec{u}\perp\vec{v}$. $\vec{u}$ and $\vec{v}$ are perpendicular to each other.